

Classifying Quadrilaterals from Their Properties

Answer Key

High School Geometry

Quick Answers

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|---|----------------------------------|
| 1. (a) $x = 36$, — | 5. — |
| 2. (a) $x = 6$, — | 6. (b) $x = 7$, — |
| 3. (a) $x = 8$, — | 7. (a) $x = 6$, — |
| 4. (a) $x = 38$, (b) larger angle = 94 , — | 8. (b) $x = 7$, (c) $y = 8$, — |
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What is verified

9 of 20 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry

HSG-CO.C — *problems 1, 2, 3, 4, 5, 6, 8*

HSG-SRT.B.5 — *problem 7*

Difficulty 1–5 of 5 across 20 tagged checks.

1. Exactly one pair of parallel sides; angles $(3x)^\circ$, $(3x)^\circ$, $(2x)^\circ$, $(2x)^\circ$. (a) Find x . (b) Most specific name, justified.

Solution (a). The four interior angles of a quadrilateral add to 360° :

$$3x + 3x + 2x + 2x = 360 \quad 10x = 360$$

so

$x = 36$

Solution (b) — **instructor-judged.** The angles are 108° , 108° , 72° and 72° . A complete response names the figure an *isosceles trapezoid*: exactly one pair of parallel sides makes it a trapezoid, and the two pairs of equal base angles make that trapezoid isosceles. Naming trapezoid alone, or naming isosceles trapezoid without citing the angle pairs, earns partial credit.

2. Diagonals of $JKLM$ cut each other into equal parts, one diagonal split into $(2x + 1)$ cm and 13 cm. (a) Find x . (b) Most specific name, justified.

Solution (a). The two parts of that diagonal are equal: $2x + 1 = 13$, so $2x = 12$ and

$x = 6$

Solution (b) — instructor-judged. A complete response names the figure a *parallelogram*: a quadrilateral whose diagonals bisect each other is a parallelogram, which is exactly what equal parts on either side of the crossing point mean. It must also note that nothing given decides whether the figure is a rectangle, rhombus or square, so parallelogram is the most specific name available.

3. Four congruent sides, one written $(3x - 4)$ cm and one 20 cm; diagonals congruent. (a) Find x . (b) Most specific name, justified.

Solution (a). All four sides are congruent, so $3x - 4 = 20$, giving $3x = 24$ and $x = 8$

Solution (b) — instructor-judged. A complete response names the figure a *square*, built from both properties: four congruent sides make it a rhombus, and congruent diagonals make a parallelogram a rectangle — a figure that is both a rhombus and a rectangle is a square. Naming square without connecting each property to the step it supplies earns partial credit.

4. $ABCD$ with both pairs of opposite angles congruent, measuring $(2x + 10)^\circ$, $(3x - 20)^\circ$, $(2x + 10)^\circ$, $(3x - 20)^\circ$. (a) Find x . (b) The larger pair of angles. (c) Classify, and give an extra condition making it a rectangle.

Solution (a). The four angles add to 360° :

$$2(2x + 10) + 2(3x - 20) = 360 \quad 10x - 20 = 360$$

so $10x = 380$ and

Solution (b). $2x + 10 = 86$ and $3x - 20 = 94$, so the larger pair measures

Solution (c) — instructor-judged. A complete response names $ABCD$ a *parallelogram*, since both pairs of opposite angles being congruent is one of the standard parallelogram tests, and then supplies a workable extra condition for a rectangle: one angle is a right angle, or all four angles are equal, or the diagonals are congruent. Both parts are needed for full credit.

5. Always, sometimes or never: (a) a rhombus is a square; (b) a square is a rhombus; (c) a trapezoid is a parallelogram.

Solution (a) — instructor-judged. *Sometimes.* A complete response says a rhombus has four congruent sides but its angles need not be right angles; a slanted rhombus is not a square, while a rhombus with right angles is one, so both outcomes occur.

Solution (b) — instructor-judged. *Always.* A complete response says a square has four congruent sides, which is the defining property of a rhombus, so every square qualifies; the extra right angles do not disqualify it.

Solution (c) — instructor-judged. *Never,* under the definition stated in the directions. A complete response says a parallelogram needs both pairs of opposite sides parallel while a trapezoid is defined here as having exactly one, so the two conditions cannot hold at once. A response that argues correctly from the inclusive definition of trapezoid may be accepted provided it states which definition it is using.

6. $EFGH$: diagonals congruent and bisecting each other, but not perpendicular; one diagonal $(4x + 5)$ cm and the other 33 cm. (a) Most specific name. (b) Find x . (c) Why can it not be a square?

Solution (a) — instructor-judged. A complete response names $EFGH$ a *rectangle*: diagonals that bisect each other make it a parallelogram, congruent diagonals make that parallelogram a rectangle, and since the diagonals are stated not to be perpendicular it is not a square — so rectangle is the most specific name.

Solution (b). The two diagonals are congruent, so $4x + 5 = 33$, giving $4x = 28$ and

$$x = 7$$

Solution (c) — instructor-judged. A complete response says a square's diagonals are perpendicular, and these are stated not to be, so the figure fails a property every square has. Citing instead that a square's diagonals bisect its angles, or that all four sides would have to be congruent, is acceptable provided the response ties the failure to a property squares must have.

7. $ABCD$ with $AB = AD = (3x - 1)$ cm, $CB = CD = 14$ cm and $AB = 17$ cm. (a) Find x . (b) Most specific name, and why it is not a rhombus.

Solution (a). $3x - 1 = 17$, so $3x = 18$ and

$$x = 6$$

Solution (b) — instructor-judged. A complete response names $ABCD$ a *kite*: its congruent sides are adjacent in two pairs (the pair at A and the pair at C) rather than opposite, and the two pairs have different lengths, 17 cm and 14 cm. A rhombus would require all four sides congruent, which fails here; that contrast must appear for full credit.

8. A student claims two pairs of congruent sides force a parallelogram. (a) Counterexample. (b) $(5x - 8)$ cm congruent to 27 cm: find x . (c) 19 cm congruent to $(2y + 3)$ cm: find y .

Solution (a) — instructor-judged. A complete response gives a *kite* and describes the arrangement: its two pairs of congruent sides are adjacent, sharing a vertex, rather than opposite each other, so no pair of opposite sides need be parallel or congruent. Any correct counterexample with an accurate description of how its congruent sides sit earns full credit; naming a shape alone does not.

Solution (b). $5x - 8 = 27$, so $5x = 35$ and

$$x = 7$$

Solution (c). $2y + 3 = 19$, so $2y = 16$ and

$$y = 8$$